

Manage Azure VM

Day-two VM operations — resizing, moving, updating, monitoring, patching, custom images, scale sets, and cost management for virtual machines already running in Azure.

What "managing" a VM means after deployment

Deploying a VM is day one. Managing it is everything after: keeping it patched, right-sized, monitored, backed up, and recoverable. These are the operations you'll perform repeatedly — and the ones the AZ-104 exam tests most heavily in the compute domain.

Eight terms you need cold

VM Resize

Changing a VM's size (CPU/RAM) to a different SKU. Requires a restart. If the new size isn't available on the current host cluster, the VM is deallocated and moved first.

Tip: Resizing within the same size family (e.g. D2s_v3 → D4s_v3) usually stays on the same host. Cross-family resizes almost always deallocate.

VM Move

Relocating a VM to a different resource group or subscription. The VM's disks, NICs, and NSGs must all move together. No downtime is required for a resource group move.

Tip: You cannot move a VM to a different region directly — use Azure Site Recovery or snapshot + redeploy for cross-region moves.

Azure Update Manager

A unified service for assessing and deploying OS updates across VMs and Arc-enabled servers. Replaces the older Update Management in Automation Accounts.

Tip: Supports scheduled patching with maintenance windows and reboot policies — critical for controlling when patching happens in production.

Scheduled Events

A metadata service endpoint inside the VM that provides advance notice (up to 15 minutes) of upcoming platform maintenance, reboot, or redeployment events.

Tip: Poll <http://169.254.169.254/metadata>

Custom Script Extension

Runs a script (PowerShell or bash) inside a VM post-deployment — the primary way to configure software, set registry keys, or install agents without manual RDP/SSH.

Tip: Only one instance of Custom Script Extension can run at a time per VM — remove the old one before adding a new script.

VM Scale Set (VMSS)

A group of identical load-balanced VMs that auto-scales based on CPU, memory, queue depth, or a schedule. Supports both Uniform and Flexible orchestration modes.

Tip: Flexible mode allows mixing VM sizes and supports Availability Zones — prefer it for new workloads.

🛡️ Disk Encryption

Two options: Azure Disk Encryption (ADE) uses BitLocker/DM-Crypt with keys in Key Vault. Server-Side Encryption (SSE) encrypts at rest at the storage layer — always on by default.

Tip: SSE is automatic and free. ADE adds OS-level encryption for compliance requirements that mandate in-guest encryption.

📄 Capture / Generalize

Sysprep (Windows) or waagent deprovision (Linux) removes machine-specific data, then the VM is captured as a reusable image in the Azure Compute Gallery.

Tip: After generalizing, the VM cannot be started — capture immediately and delete the source VM.

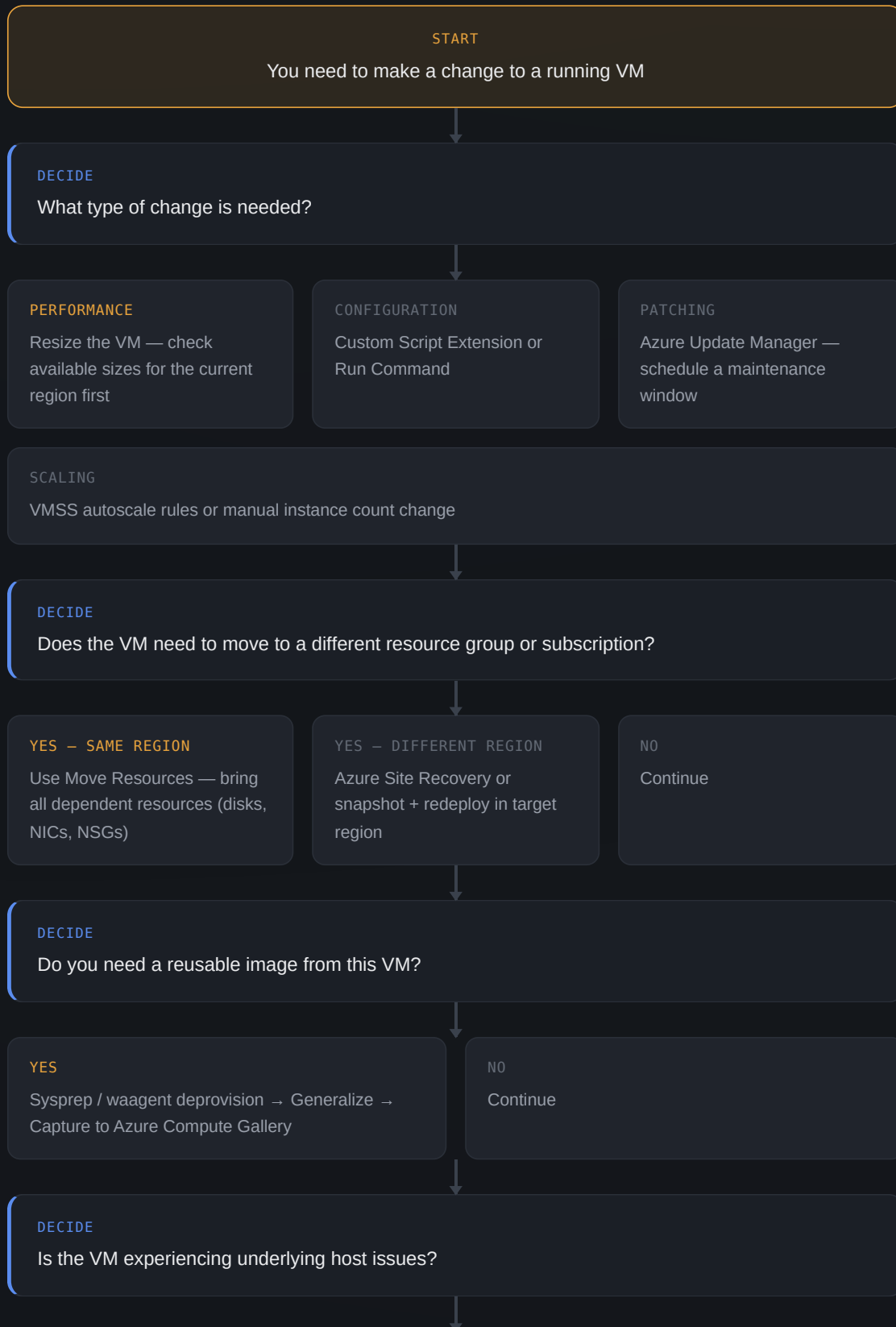
OPERATION	DOWNTIME?	LOSES DYNAMIC PUBLIC IP?	NOTES
Restart	Brief	No	Stays on same host
Stop (deallocate)	Yes	Yes	Compute billing stops
Resize (same family)	Brief restart	Usually no	May stay on same host
Resize (cross-family)	Restart + dealloc	Yes	Host reallocation likely
Move (resource group)	No	No	All dependent resources move together
Redeploy	Yes	Yes	Moves VM to a new host — troubleshooting tool

EXAM GOTCHA — REDEPLOY VS RESTART

Restart reboots the OS on the same host — fast, no IP change. **Redeploy** moves the VM to a completely different physical host and then starts it — it loses its dynamic public IP and is only used to resolve underlying host hardware issues, not as a routine restart.

Day-two VM management decision flow

Use this when you need to change, fix, or maintain an existing VM.



YES – HOST PROBLEM

Use Redeploy to move to a new host (accepts
downtime + IP loss)

NO – OS / APP ISSUE

Use Run Command or Serial Console for in-guest
diagnosis



DONE

VM updated, moved, imaged, or recovered as needed

Tools for day-two VM management

> Azure CLI — lifecycle ops

BEST FOR: scripted `resize`, `stop`, `start`, `redeploy`

Fast one-liners for all common VM lifecycle operations — easy to chain in maintenance scripts.

```
az vm resize -g rg-prod -n vm-web01 \
  --size Standard_D4s_v3

az vm restart -g rg-prod -n vm-web01

az vm redeploy -g rg-prod -n vm-web01

az vm list-vm-resize-options \
  -g rg-prod -n vm-web01 \
  --query "[].name" -o table
```

> Azure CLI — extensions & images

BEST FOR: post-deploy config, image capture

Apply extensions, run commands, and capture images without touching the VM via RDP/SSH.

```
az vm extension set \
  -g rg-prod -n vm-web01 \
  --extension-instance-name config \
  --publisher Microsoft.Azure.Extensions \
  --name CustomScript \
  --settings '{"fileUris":
["https://.../setup.sh"],"commandToExecute":"b
ash setup.sh"}'

az vm generalize -g rg-prod -n vm-web01
az image create -g rg-prod -n img-web \
  --source vm-web01
```

☒ Azure Portal — Update Manager

BEST FOR: patch scheduling, compliance assessment

Azure Update Manager gives a visual compliance dashboard and lets you schedule maintenance windows with reboot policies — far safer than manual patching.

```
Azure Update Manager → Machines
→ Schedule updates → Set maintenance window
→ Choose reboot policy
→ Review + create schedule
```

> Azure CLI — VMSS

BEST FOR: scaling, instance management

Scale instances manually or update autoscale rules from the command line — useful in pipeline scripts that respond to deployment events.

```
az vmss scale -g rg-prod \
  -n vmss-web --new-capacity 5

az monitor autoscale create \
  -g rg-prod --resource vmss-web \
  --resource-type
Microsoft.Compute/virtualMachineScaleSets \
  --min-count 2 --max-count 10 --count 2
```



Run Command / Serial Console

BEST FOR: emergency access when SSH/RDP is unavailable

Run Command executes scripts via the VM agent without network connectivity. Serial Console provides a text console at the BIOS/boot level — the last resort before reimaging.

```
az vm run-command invoke \
  -g rg-prod -n vm-web01 \
  --command-id RunShellScript \
  --scripts "systemctl status nginx"
```

```
# Serial Console: Portal → VM
#   → Help → Serial console
```

TOOL	BEST FOR	NEEDS NETWORK TO VM?
CLI / PowerShell	Lifecycle ops, extensions, scaling	No — control plane
Azure Bastion	Interactive RDP/SSH sessions	Yes — via VNet
Run Command	Script execution without network path	No — via VM agent
Serial Console	Boot-level / OS recovery	No — out-of-band
Update Manager	OS patching, compliance scheduling	No — via Azure agent

Principles worth internalizing

ENCRYPTION LAYERS — KNOW THE DIFFERENCE

Server-Side Encryption (SSE) is always on — Azure encrypts all managed disk data at rest at the storage layer for free. **Azure Disk Encryption (ADE)** adds in-guest OS-level encryption (BitLocker for Windows, DM-Crypt for Linux) using keys stored in Key Vault — required by compliance frameworks that mandate guest-level encryption. You need both only for the strictest compliance requirements.

EXAM GOTCHA — CROSS-REGION VM MOVE

There is **no direct "move to a different region"** button for VMs. Options are: Azure Site Recovery (replicates the VM continuously and allows a planned failover), or manually taking a snapshot, copying the disk to the target region, and deploying a new VM from it. The exam tests that you know cross-region moves require one of these approaches.

Six mistakes that show up on the exam

Using Redeploy as a routine restart

Redeploy moves the VM to a new host, loses the dynamic public IP, and causes real downtime — it's a troubleshooting tool for host hardware issues, not a restart.

Forgetting to check available sizes before resize

Not all VM sizes are available in every region or on every host cluster. Always run `az vm list-vm-resize-options` first — a resize to an unavailable size fails with a confusing error.

Running Custom Script Extension twice on the same VM

Only one instance of Custom Script Extension can exist per VM — attempting to add a second fails. Remove the existing extension first, then apply the new one.

Trying to move a VM to a different region via resource move

The Move Resources function supports cross-subscription and cross-resource-group, but not cross-region. Use Azure Site Recovery or snapshot + redeploy for region moves.

Confusing SSE and ADE

SSE is automatic and always on (storage layer). ADE is an additional in-guest encryption layer for compliance — you enable ADE on top of SSE, not instead of it.

Starting a generalized VM

After Sysprep/waagent deprovision, the VM is non-bootable and will corrupt the image if started. Capture the image immediately after generalizing, then delete the source VM.